**UNIVERSIDAD DE LAS FUERZAS ARMADAS ESPE**
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# TEST-DRIVEN DEVELOPMENT (TDD)

Better code. Fewer bugs. Greater confidence.

## WHAT IS TDD?

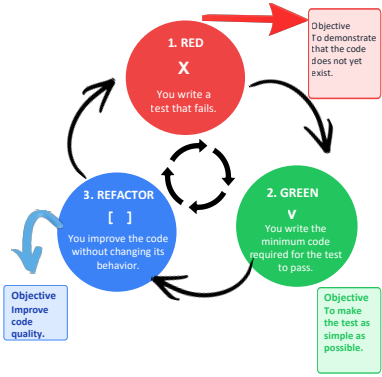
**Test-Driven Development (TDD)**
is a software development technique where the tests are written first, and then the code needed for those tests to pass is written(IBM. 2024).





Key idea: testing guides development.

## THE TDD CYCLE




## ADVANTAGES

- **Fewer errors**
  Tests detect faults early.
- **Cleaner, more maintainable code**
  Refactoring improves code design.
- **Faster long-term development**
  Less time spent fixing bugs.
- **Confidence when modifying**
  You can change the code without fear.

## DISADVANTAGES

- **Learning curve**
  Requires practice and discipline.
- **Longer at the beginning**
  It may seem slower at first.
- **It's not always appropriate.**
  Not all projects need it.
- **Dependence on good tests**
  Poorly done tests = false security.

## SIMPLE EXAMPLE

**1. NETWORK - Test (failure)**

```
test('sumar 2 + 2 debe
ser 4', () => {
  expect(sumar(2,
2)).toBe(4);
});
```

The test fails because the function does not exist.

**2. GREEN - Minimum code**

```
function sumar(a, b) {
  return a + b;
}
```

The test passes.

**3. REFACTOR - Improvement**

```
function sumar(a, b) {
  const resultado = a +
b;
  return resultado;
}
```

Improvements without changing the outcome.

*"First the test, then the code."*

**LITERATURE**

IBM. (2024). Test-driven development (TDD). https://www.ibm.com/es-es/think/topics/test-driven-development
SoftwareCrafters. (s.f.). Test-driven development en JavaScript. https://softwarecrafters.io/javascript/tdd-test-driven-development
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